

PRILEP, North Macedonia

WMO: 135850

Lat: 41.333N

Lon: 21.567E

Elev: 2211

StdP: 13.56

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	17.4	6.2	8.2	14.8	11.0	10.4	21.5	31.4	45.3	25.0	38.9	1.6	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	92.7	66.4	89.4	65.6	86.5	64.7	68.6	85.8	67.4	84.0	66.2	82.2	5.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	96.4	71.1	62.4	91.6	70.7	61.0	87.0	70.3	34.0	86.3	33.0	83.9	32.0	82.4	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.1	15.6	6.8	97.5	6.2	4.1	2.4	100.4	-1.3	102.8	-4.8	105.1	-9.3	108.1
			6.1	70.7	6.4	1.4	1.5	71.7	-2.2	72.5	-5.8	73.3	-10.4	74.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	33.1	38.2	45.7	53.2	60.7	68.2	73.4	73.6	65.1	55.7	46.8	36.0	(6)
	DBStd	15.34	8.77	8.45	6.75	6.53	5.56	5.65	5.26	4.66	5.81	6.77	7.01	8.80	(7)
	HDD50	1673	523	336	167	39	1	0	0	0	0	30	139	438	(8)
	HDD65	4691	988	751	599	356	155	28	5	1	70	294	547	897	(9)
	CDD50	3214	1	4	33	134	332	547	726	731	452	206	42	6	(10)
	CDD65	758	0	0	0	1	21	126	266	268	72	4	0	0	(11)
	CDH74	7429	0	0	0	29	232	1225	2723	2559	606	55	0	0	(12)
	CDH80	2775	0	0	0	1	42	403	1132	1033	159	5	0	0	(13)
(14) Wind	WSAvg	4.1	3.6	4.3	5.9	4.9	4.1	4.3	4.5	4.0	3.8	3.3	3.4	3.4	(14)
(15) Precipitation	PrecAvg	22.8	1.7	1.7	1.8	1.9	2.8	2.0	1.3	1.3	1.7	2.0	2.6	2.0	(15)
	PrecMax	29.6	4.2	4.9	4.2	4.3	4.8	5.0	3.5	4.6	4.1	5.3	8.9	4.1	(16)
	PrecMin	16.3	0.6	0.2	0.3	0.9	0.3	0.4	0.0	0.0	0.1	0.0	0.5	0.2	(17)
	PrecStd	3.8	0.9	1.2	1.0	0.9	1.3	1.4	1.2	1.0	1.2	1.4	2.1	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	64.2	69.4	77.9	85.0	93.3	97.4	96.4	88.7	79.6	68.9	61.4	(19)
		MCWB	46.1	48.9	51.0	57.0	62.2	66.5	67.5	67.3	64.2	60.6	55.2	51.5	(20)
	2%	DB	51.7	57.5	65.4	72.6	79.8	88.3	93.2	92.7	84.8	74.7	64.5	55.1	(21)
		MCWB	43.8	46.2	49.3	55.2	60.7	65.5	66.3	66.5	62.7	58.0	53.6	48.3	(22)
	5%	DB	48.2	53.7	61.9	68.7	76.5	84.9	89.7	89.2	80.8	71.4	60.8	51.0	(23)
		MCWB	42.3	44.7	47.9	53.5	59.1	64.2	65.6	65.7	61.3	56.4	51.2	45.9	(24)
	10%	DB	44.9	50.2	58.1	65.1	73.0	81.6	86.5	86.0	77.3	67.4	57.4	47.6	(25)
		MCWB	40.7	42.8	46.1	51.7	57.9	63.1	64.9	64.7	60.0	55.5	50.3	44.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	50.8	52.7	59.1	65.0	68.8	70.0	70.8	66.6	61.8	57.6	52.2	(27)
		MCDB	53.1	59.7	63.9	73.2	79.1	85.1	87.9	88.7	82.1	74.2	64.6	58.9	(28)
	2%	WB	44.8	48.1	50.6	56.8	62.3	67.2	68.5	68.8	64.4	60.0	54.7	49.3	(29)
		MCDB	49.5	55.8	62.6	68.5	75.2	83.6	85.7	86.4	78.4	70.0	60.6	53.1	(30)
	5%	WB	43.0	45.7	48.8	55.0	60.7	65.7	67.4	67.2	62.9	58.5	53.0	47.0	(31)
		MCDB	47.7	52.2	59.5	65.4	72.5	81.1	84.1	84.1	75.9	66.7	58.8	50.4	(32)
	10%	WB	40.8	43.6	47.0	53.2	59.2	64.3	66.3	65.9	61.6	57.0	51.1	44.5	(33)
		MCDB	44.4	49.2	56.4	62.3	70.1	78.0	82.4	81.9	73.0	64.7	56.2	47.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	14.8	16.4	17.5	18.6	20.6	22.2	22.5	19.5	17.8	14.8	11.5	(35)
		MCDBR	17.1	21.5	24.4	23.9	24.9	26.1	28.0	27.0	25.6	24.7	20.8	15.6	(36)
	5% WB	MCWBR	10.5	11.5	12.4	10.7	9.7	9.3	10.0	9.8	9.1	10.4	11.0	10.0	(37)
		MCWBR	15.2	19.1	21.6	20.5	21.7	24.1	24.0	24.3	21.3	19.3	17.0	12.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.320	0.347	0.392	0.448	0.441	0.420	0.421	0.437	0.421	0.382	0.349	0.323	(40)
		taud	2.375	2.292	2.199	2.050	2.137	2.243	2.236	2.188	2.209	2.305	2.360	2.366	(41)
	Ebn at Noon		261	270	269	261	266	272	270	262	257	254	247	249	(42)
		Edh at Noon	26	33	41	51	48	43	43	44	40	33	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		542	819	1146	1481	1796	2067	2166	1931	1410	987	636	469	(44)
	RadStd		81	101	139	170	114	153	124	120	119	110	65	83	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-225	N/A	N/A

Nomenclature: See separate page